BT **BREATHETWIN**

TEAM MOMENTUM

*EARLY WARNING. EASIER BREATHING.*

# How the twin works: three data streams, one personal model

## Data In

**Health record (EHR)**
*400 synthetic patients: Synthea + India overlay*

**Wearable + smart inhaler**
*heart rate, breathing, SpO2, steps, sleep, puffs, doses*

**City air + weather**
*real CPCB PM2.5 and Open-Meteo, 8 cities*

## The Twin

**1. Personal baseline**
Every reading compared with the patient's **own last 4 weeks**, not a textbook range

**2. Hybrid risk engine**
**LightGBM** learns body signals, inhaler use, weather and record × **published PM2.5 effect** (DeVries 2017)

## What Comes Out

- **Risk of a flare-up in the next 3 days**
- **Reasons, ranked by how much they push risk**
- **Safety checks against own normal**
- **What-if answers for doctor and family**
- **Smog warnings + one-page visit summary**

## Twin journey

**Health record** → **Daily readings** → **Own normal** → **Risk score** → **Reasons** → **Action + advice**

*Why hybrid: the body-signal patterns are complex and best learned from data; the air-pollution effect is small and well measured in large studies, so we take it from the literature.*